Multimodal Human Activity Recognition using Fusion Strategies

Miguel Salinas Gancedo [1] Antonio López [1] José María Enguita [1]

[1]Universidad Oviedo

Abstract

Wearable inertial sensors have become essential for digital health monitoring, yet most large-scale cohort studies still rely on single-sensor placements, typically at the wrist or thigh. While both locations offer unique advantages, single-sensor setups often fail to capture the full complexity of human movement. This study systematically evaluates multimodal fusion strategies for integrating data from wrist and thigh-mounted Inertial Measurement Units (IMUs) to improve physical activity classification. Using data from the WearablePerMed project which mirrors the dual-sensor protocol of the IMPACT (Spain) cohort we compared single-sensor baselines against early fusion (feature concatenation) and late fusion (Stacking and Mixture of Experts) approaches across three levels of activity granularity (4, 8, and 15 classes).

Introduction

This study evaluates multimodal fusion of wrist and thigh IMUs to improve physical activity classification, overcoming the limitations of common single-sensor setups that fail to capture complex human movement.

- **Sensor Complementarity:** Wrist devices excel at upper limb dynamics but are suboptimal for posture, while thigh sensors accurately estimate sitting and standing but lack sensitivity to fine arm movements.
- **Methodological Gap:** Despite the growth of multisensor initiatives like the IMPACT cohort, there is a lack of comparative evidence on which fusion strategies—ranging from simple concatenation to Mixture-of-Experts provide the most reliable gains.
- **Research Objective:** Using data from the WearablePerMed project, this work quantifies the performance of various integration strategies across multiple activity granularities to establish evidence-based guidelines for future multisite sensor deployments.

This work provides new methodological insights into the design of multisite wearable-sensor systems to achieve more accurate and comprehensive characterization of human behavior.

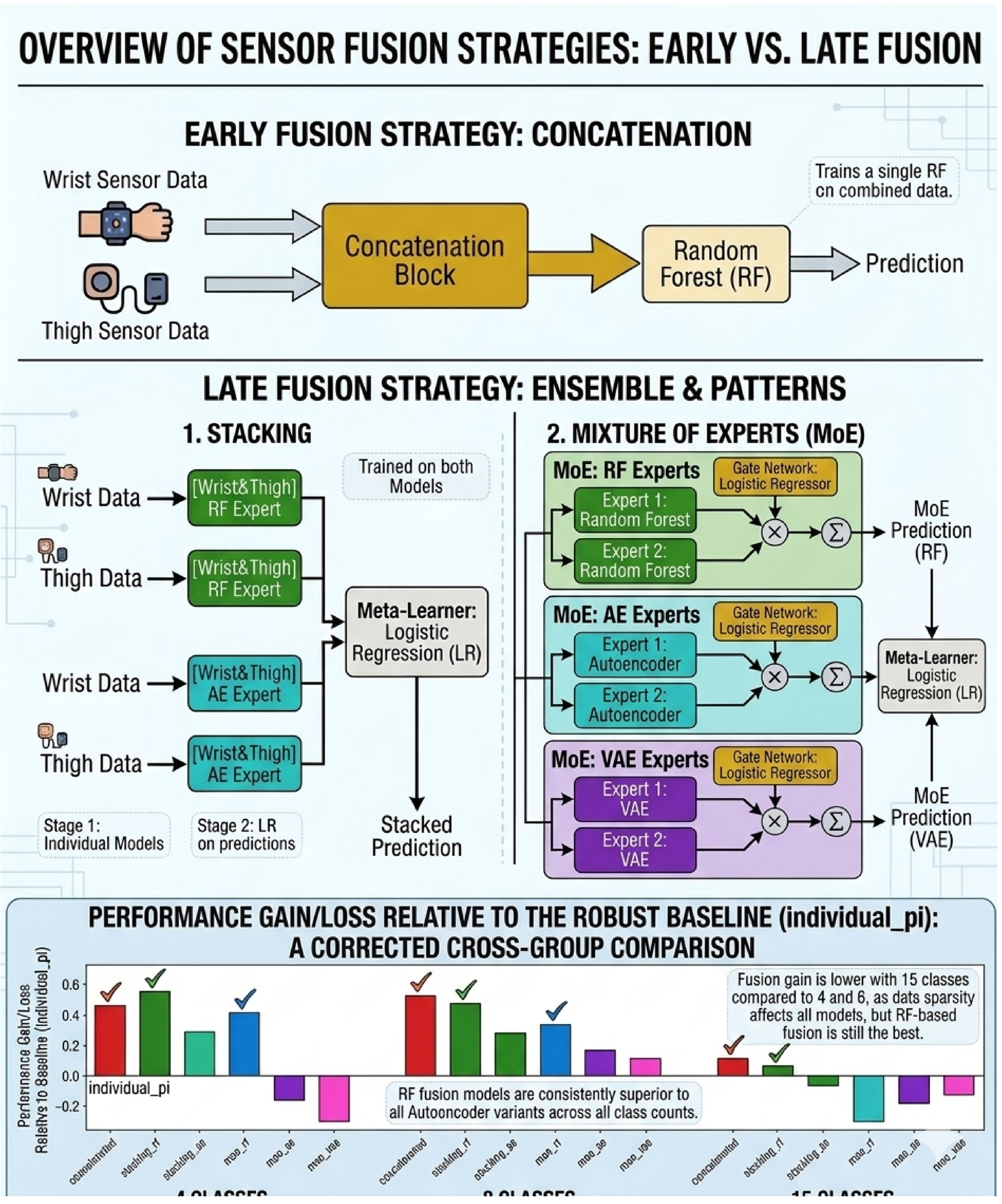
Fusion and Method Strategies

This section details sensor fusion architectures for classification. Early Fusion uses raw signal concatenation to train a single Random Forest (RF) model. Late Fusion, via Stacking and MoE patterns with RF and AE/VAE variants, employs individual models to feed a final Logistic Regression meta-learner.

Table 1. Key Sensor Fusion Strategy Comparisons

Fusion Strategy	Method	Performance Observation
Early Fusion	Concatenation	Robust with fewer classes.
Late Fusion: RF	Stacking/MoE	Best performance across all class counts.
Late Fusion: AE/VAE	Stacking/MoE	Lower accuracy; scales poorly with 15 classes.

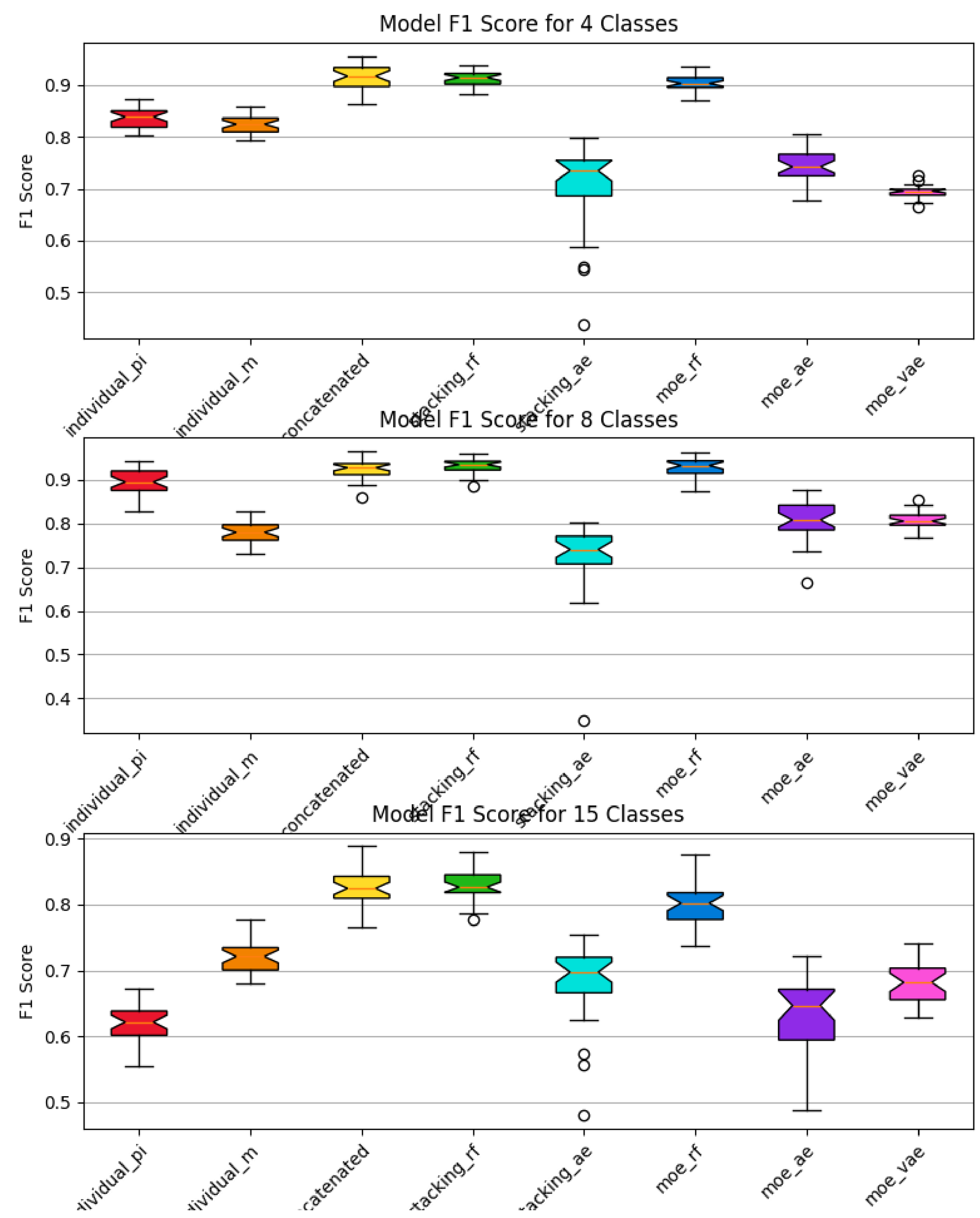
Figure 1. Fusion Models Methods



Results

1. **Fusion improves performance:** both early fusion (feature concatenation) and late fusion (stacking and MoE) outperform individual sensor models, supporting the hypothesis that multi-sensor information is complementary.
2. **Late fusion is superior to early fusion:** stacking and mixture-of-experts approaches generally achieve higher median F1-scores than simple concatenation.
3. **Random Forest–based fusion is the most effective:** models using Random Forest as base learners or experts consistently yield the highest performance and stability.
4. **Representation learning approaches show higher variance:** autoencoder and VAE-based models underperform compared to tree-based methods, particularly in more complex classification scenarios.
5. **Performance decreases with class granularity, but fusion methods remain advantageous:** although performance drops as the number of classes increases, fusion methods maintain a consistent advantage even in the most challenging 15-class setting.

Figure 2. F1 Score (%) for different fusion strategies and granularities



Future Work

1. Web portal for Human Activity Recognition (HAR) to predict user activities.
2. Scalable application that supports uploading CSV files with accelerometer and gyroscope data.
3. Secure platform implementing OAuth2 for authentication and authorization.
4. Multitenant architecture enabling isolated environments for different users or organizations.
5. Support for multiple predictive models to improve accuracy and robustness.

Figure 3. Predictor view from WearablePerMed Portal